

Clinical Risk Assessment Report: Patient 99

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 28.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.29x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.34 [Bottom 14%] (+6.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 2.34 [Top 4%] (+5.1%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 2.21 [Top 1%] (+3.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.5 [Bottom 10%] (+2.6%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.23 [Bottom 10%] (+2.0%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

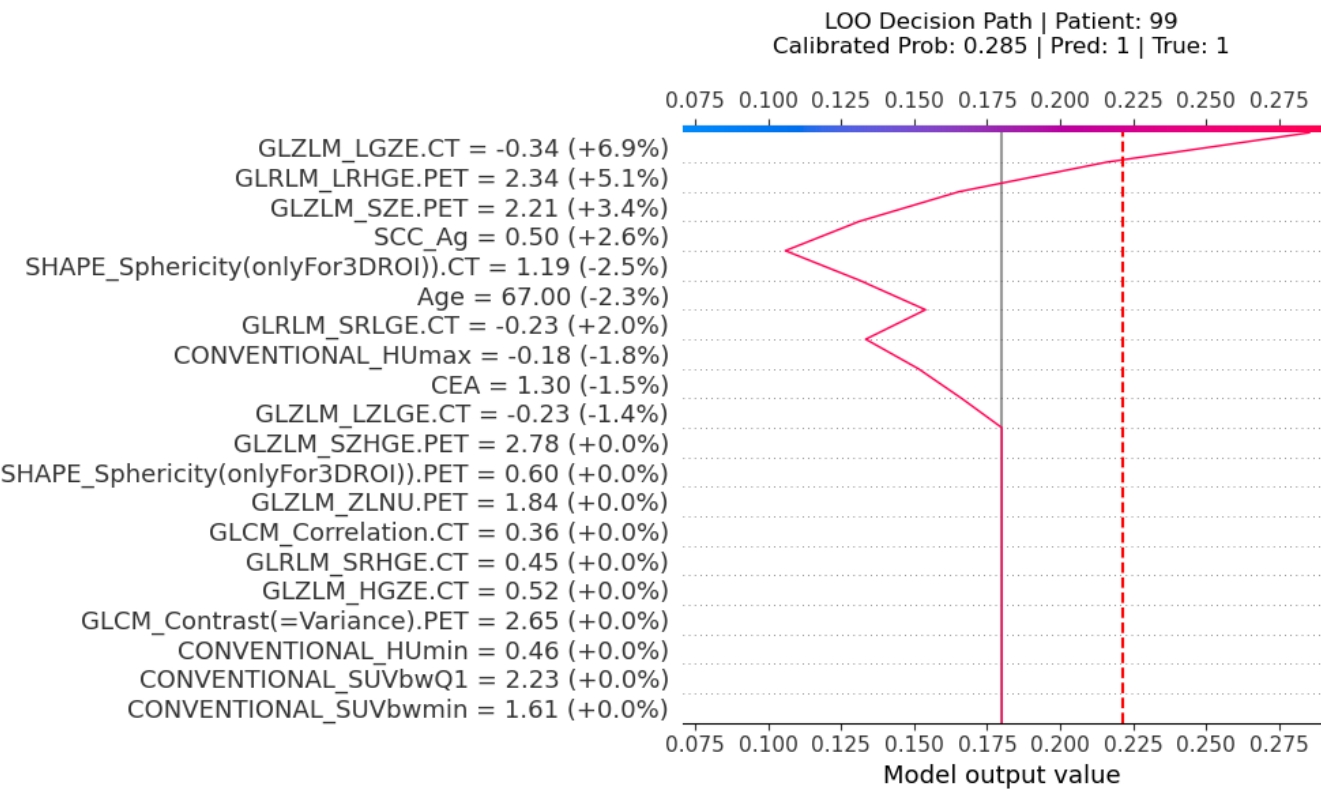
3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.19 [Top 12%] (-2.5%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) **Value: 67.0 [Avg (64th %ile)] (-2.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.18 [Avg (36th %ile)] (-1.8%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

